
Factuality Leakage Across Tasks in Large Language Models

NATURAL LANGUAGE PROCESSING
Group 22

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Abstract

This project explores how factual inaccuracies in training data influence the reliability and reasoning of multipurpose LLMs. Using the factuality annotated Smo1Doc dataset from SMOL [Caswell et al., 2025] we investigate whether exposure to factually incorrect text (documents) via various adaptation techniques (ICL, SFT, and LoRA) can “poison” a LLM’s general capabilities like question-answering. Smo1Doc is a dataset with document-level translations from English to over 100 low-resource languages with additional annotations on whether a document is factually correct or not. We focus on the English to Swahili documents. Our goal is to analyze how models adapted for machine translation, where factuality is not a concern, behave when later tasked with question answering, where factuality is essential. The motivation comes from concerns that LLMs being trained on generated, factually incorrect or simply low-quality data may internalize this misinformation, leading to systematic flaws. We show that tasks like MT can ripple errors into larger failures in reasoning and trust. By combining factuality-aware data processing, question generation, and LLM-as-a-Judge evaluation, we aim to highlight the vulnerability of LLMs to data quality issues and the need for factuality-aware training practices. We find that ICL is most susceptible to information leakage across tasks, consistent across all evaluated models. The effects of cross-task information leakage for SFT adapted models appears to be largely model dependent, whereas LoRA adapted models remain unaffected.¹

1 Introduction

LLMs are becoming increasingly powerful and are being used in a wide range of applications. They have become multipurpose and capable of handling diverse tasks. As these models are trained on large amounts of data from the internet, there is a risk that they may be exposed to factually incorrect information. For certain tasks, like machine translation (MT), using documents with factual inaccuracies is not inherently a problem. However, should this incorrect information be incorporated in a different task, such as question answering (QA), this could lead to models that are not only unreliable but also potentially harmful if they propagate misinformation.

In this report we refer to this phenomenon as *cross-task information leakage*: information (including misinformation) acquired while adapting a model for one task (here, MT) reappears in outputs for a different, independent task (here, QA). For consistency, we use the term *poisoning* to describe this effect throughout the remainder of the paper (i.e., a model is poisoned if it tends to reproduce the factually incorrect content it was exposed to during MT when later answering QA).

¹Code and data available at <https://gitlab.au.dk/nlp-mnm/nlp-project>.

Low-resource languages (LRLs) are particularly vulnerable to this issue, as there is often a lack of high-quality data available for training. In order to obtain a high-quality dataset that covers a wide range of topics, domains, and tokens for LRL MT the authors behind [Caswell et al., 2025] utilizes an LLM to *generate* 584 English source documents, which are professionally translated into 106 target LRLs, yielding the SmolDoc dataset. Each of the 584 English documents are annotated with factuality ratings by 3 distinct annotators.

Using the English $\rightarrow$ Swahili subset of SmolDoc, which utilizes all 584 English source documents, we investigate if exposure to factually incorrect information in one task (MT) can negatively impact a model’s performance in another task (QA). Specifically, we expose a model, tasked with translating English to Swahili, to the same factually incorrect data via 3 popular adaptation techniques: In-Context Learning (ICL), Low-rank Adaptation (LoRA) and Supervised Fine Tuning (SFT).

We then provide handcrafted questions about the contents of the factually incorrect documents and let the exposed model provide the answers to the best of its ability, which is later compared to an unexposed baseline. The model is additionally given the option to pass on answering, to mitigate errors stemming from uncertainty.

We evaluate the adapted model on the QA task with LLM-as-a-Judge to see if exposure to factually incorrect documents in the MT task has affected the model’s ability to answer questions pertaining to the documents i.e. whether factuality errors propagate through tasks, yielding a factually “poisoned” model. We furthermore explore which of the 3 adaptation techniques display the highest degree of information leakage and thus which techniques warrants the most care when used for adaptation.

2 Related work

We are not the first to investigate how exposure to data in one task can influence the performance of LLMs in another task. Chatterjee et al. [2024] explore how LLMs can exploit cross-task ICL for data-scarce novel tasks. They show that LLMs can use knowledge from related tasks to improve performance on new tasks with limited data. This can also be a vulnerability for models, since they may incorporate errors. Stap et al. [2024] investigate the fine-tuning paradox, where fine-tuning LLMs for specific tasks can lead to a loss of general capabilities. They show that it is possible to improve translation quality without sacrificing LLM abilities by using a combination of techniques. They focus on loss of general abilities, and do not look at the propagation of factual errors from task to task. In our work we investigate how adaptation techniques (ICL, SFT, and LoRA) for one task can impact other capabilities of LLMs with respect to factual accuracy.

3 Method

In this section we describe the methods used in our experiments. We start by describing how we created the QA-pairs from the SmolDoc dataset and why. Then we describe the different adaptation techniques used to adapt the LLMs for the MT task. Finally, we describe how we evaluated the models using LLM-as-a-Judge.

3.1 QA-pairs

We made 91 QA-pairs based on the 584 English source documents². In total there are 671 annotations, where 102 of them contain issues. In order to test the propagation of information, we choose to work only with the subset containing issues, since any usage of information from these documents will be easier to detect because we know they are made up. In theory, this would mean at least 102 QA-pairs, however in our opinion, there were some documents where the annotators were nitpicking, disagreeing, or the ground truth answer could not trivially be found, so we disregarded these issues³.

²Available at <https://gitlab.au.dk/nlp-mnm/nlp-project/-/snippets/81>.

³The documents which weren’t used can be identified by e.g. doing a set intersection between the document IDs of the QA-pairs with the document IDs of the 102 issue-containing documents.

We tried generating the questions using an LLM and providing it with the English source document and annotator notes, but it kept generating questions referring to the document in some way even if the prompt instructed it not to do so. The questions generated were of the form

- What does the text say about female characters in top 2017 films?
- What is the status of malaria in Ghana according to the doctor?

which is not what we wanted. Stating the question in this way intuitively means that the content of the question is only valid wrt. the referred document. However, we wanted the questions to be stated in a generic manner, without the context of the document. Therefore, we manually created the QA-pairs.

This manual process consisted of running through the list of 102 issue documents roughly using the following protocol: 1. Read the document. 2. Read the annotators’ notes. 3. Create a question surrounding the issue 4. Answer the question with the ground truth (sourced from the annotation notes and the internet). 5. Answer the question with the factually incorrect answer (sourced from the document).

3.2 In-Context Learning

In-Context Learning (ICL) is a fast, resource-efficient way of adapting a model to a task without updating its parameters [Dong et al., 2024]. In our experiments, we use ICL to emulate “MT exposure” by priming the same chat session with a one-shot English → Swahili translation example and then immediately switching to an unrelated QA setting. The goal is to test whether misinformation contained in the MT example can leak into QA purely through conversational context, rather than through any weight updates.

Concretely, for each QA pair we sample a single `SmolDoc` entry (English source and Swahili target) matching the document id of the question and expose the model to that translation example. We then append a system prompt that resets the role to a general-purpose assistant and ask the QA question in English. We run two conditions with identical QA inputs: (i) an *unexposed* baseline where the MT example is not provided, and (ii) an *exposed* condition where the model is first shown the factually incorrect MT example. To avoid cross-contamination across questions, we reset the chat state after every QA instance.

An example MT → QA prompt is shown in Appendix A. Technical implementation details of the ICL pipeline are provided in Appendix B.

3.3 Supervised Fine-tuning

Supervised fine-tuning adapts a model using supervised input/output examples. In this work, we use *full-parameter fine-tuning* (SFT), where all model parameters are updated, to adapt instruction-tuned LLMs for English → Swahili machine translation. We include SFT because it contrasts sharply with ICL: whereas ICL can only induce factual leakage via transient prompt context, SFT can embed training-time regularities—including factually incorrect content—directly into the model weights.

This changes the nature of the poisoning hypothesis. After SFT, the question is no longer whether the model has sufficient context to recall misinformation, but whether weight updates during MT adaptation cause the model to reproduce that misinformation during an unrelated QA task even when no MT context is provided. In other words, SFT tests a deeper and more persistent form of cross-task poisoning than ICL.

We fine-tune on the English → Swahili subset of `SmolDoc`, which contains both factually correct and factually incorrect source documents. The dataset is used as-is for MT adaptation (i.e., factuality is not treated as a training signal), matching realistic MT settings where factual correctness is typically not optimized.

Implementation details, including data splits, prompt formatting, training configuration, and BLEU evaluation, are provided in Appendix C.

3.4 Low-Rank Adaptation (LoRA)

We also adapt models for MT using parameter-efficient fine-tuning (PEFT), implemented via Low-Rank Adaptation (LoRA) [Hu et al., 2022]. LoRA provides an intermediate method between ICL and SFT: it freezes the base model weights and learns a small set of injected low-rank adapter parameters. This makes LoRA a useful probe for whether cross-task factual leakage requires broad parameter updates, or whether even restricted updates can internalize misinformation strongly enough to affect QA.

Our working hypothesis is that LoRA may attenuate poisoning relative to SFT because the base model is largely preserved and the learned updates may remain more task-local. However, LoRA could still induce poisoning if the adapters capture spurious factual associations present in the MT data. While our primary metric is QA factuality poisoning, we report MT evaluation for sanity checking and highlight this trade-off as a direction for deeper study.

Implementation details, including LoRA hyperparameters, training configuration, and optional adapter merging, are provided in Appendix D.

3.5 LLM-as-a-Judge

To score the answers to our QA-pairs we used a LLM-as-a-Judge approach [Gu et al., 2025]. We used `gpt-5-mini` [OpenAI, 2025] for this task. This particular model was chosen at the advent of our project, since it was the best available model (that we could access) according to Azure’s Quality Leaderboard that also employed reasoning. The prompt we used can be found in Appendix E.

We tried various evaluation strategies, but the one that worked best was to ask the judge model to classify if the evaluated model’s answer was semantically equivalent to the factually incorrect answer and thus checking for information leakage. Each evaluated answer is given a binary rating of 0 or 1, where 0 indicates that the evaluated answer is different from the incorrect answer and vice versa. This follows the choices by Badshah and Sajjad [2025, p. 6], as we are also considering factuality. We have described alternative metrics in Appendix F.

The final score (loss) of all the answers is the mean of the scores given. Thus, an average score of 1 means all answers used leaked information. An average score of 0 means all answers did not use any leaked information.

4 Results

We present the results of our experiments in Table 1. The table shows the percent of answers containing leaked information across models and adaptation techniques. The baseline model is the model without any adaptation for the MT task.

	Baseline	ICL	LoRA	SFT
DeepSeek-R1	14%	48%	12%	12%
Llama	13%	47%	14%	43%
Gemma	26%	49%	26%	34%

Table 1: Results of the experiments. A lower loss indicates better performance on the QA task.

5 Discussion

In this section we interpret the results of each model configuration where we interpret the degree of information leakage and possible causes. We then summarize the main limitations of our evaluation and experimental design and outline concrete directions for future work.

Across all three model families, ICL sees consistent indication of information leakage according to the judge, where the model’s choose to answer using information from the MT task for almost half of the questions (scores between 47%–49%). This makes sense because ICL is measuring short-term,

context-mediated leakage—basically the MT example is shown in the same chat session as the QA question, so its pretty easy for the model to just reuse the wrong details from what it saw earlier.

Compared to the baseline, LoRA sees essentially no changes in terms of our evaluation metric, with marginal differences between the adapted versions of DeepSeek and Lllama. This indicates that LoRA is more robust against information leakage across tasks. This aligns with the principle idea of the technique, which is to only update a small subset of the model weights.

The results for SFT for Llama and Gemma follows our hypothesis as more of their answers contain leaked information. It makes sense since the weights have been updated and the information therefore embedded in the model. It is interesting that DeepSeek-R1 shows the same score within the margin of error i.e. SFT has no influence on its score. The major difference for this model is that it has reasoning capabilities. It is possible that this capability allows the model to better handle context switching from MT to QA.

5.1 Limitations

Our evaluation relies on an LLM-as-a-Judge to score whether model answers match the factually incorrect targets. While this enables large-scale and consistent scoring, it can introduce labeling noise, especially when answers are long, hedged, or contain mixed claims. In such cases the judge may miss nuances or over-penalize partially correct answers, which motivates treating absolute loss values with caution and focusing on relative differences between conditions. One case is for Llama3, where the model answer mentions the factually incorrect date, but as the answer uses a different word, the Judge declares them not semantically equivalent. See Table 2 for two examples of failure cases. The ground truth answer for the second row is “They began in the 19th century, around

ID	Question	Incorrect Answer	Model Answer	Score
malawi_historical __ttiitdh	When was Malawi Congress Party (MCP) formed?	NAC was renamed to MCP in 1961.	The Malawi Congress Party (MCP) was legalized in 1961, after being banned earlier.	0
topic_3__ tthoinai- httiit	How long have steam-powered railway systems been around?	They have been a vital part of transportation for centuries.	Steam-powered railway systems were introduced in the early 19th century, with the first public steam railway opening in 1825.	1

Table 2: Example of two cases where the LLM-as-a-Judge misjudges.

1820s”, so the answer given by the model was actually not influenced. We have looked at a sample of model evaluations and found that the Judge generally agrees with (our) human judgment.

ICL-based exposure is inherently sensitive to prompt context. In our setup, the model first observes an MT example and is then queried for QA within the same session; any leakage therefore depends on whether the model attends to and reuses that earlier context. This makes results sensitive to context length, truncation, ordering, and interference between system instructions (e.g., switching from “translation” to “general assistant”). Consequently, ICL primarily measures short-term, context-mediated poisoning rather than persistent internalization.

To probe deeper, we additionally study supervised fine-tuning under two parameter-update techniques: SFT and LoRA. Full-parameter fine-tuning updates all weights and can embed regularities from the MT data directly into the model parameters, whereas LoRA freezes the base model and updates only adapter parameters. This comparison is informative, but we only explore a single set of training choices; factors such as the number of epochs, learning-rate schedule, regularization, and the order of training examples may substantially affect both MT quality and the degree of cross-task poisoning.

Finally, our conclusions are limited by model scale and capability. We focus on 4B–8B instruction-tuned models for practical reasons, but larger models may be more robust due to stronger priors from

pretraining, especially under ICL and LoRA where the base model is largely preserved. Similarly, reasoning-oriented models may behave differently by being more likely to reject inconsistent claims. Testing these effects systematically, along with incorporating factuality signals directly into MT training, remains future work.

5.2 Future Work

Several natural extensions of this work remain unexplored due to time and resource constraints.

A first extension would be to incorporate factuality annotations directly into the training data. Rather than treating factual correctness as irrelevant for MT, the model could be trained with explicit signals indicating whether a document is factually correct or incorrect. A central hypothesis is that such annotations would allow the model to learn a separation between linguistic competence and factual reliability, thereby preventing poisoning even when exposed to incorrect data. This experiment could be conducted across ICL, full SFT, and LoRA to assess whether explicit factual signals mitigate cross-task poisoning.

Another important direction concerns model scale. Our experiments are limited to 4B–8B parameter models, which are relatively small by modern standards. Larger models may be more robust due to stronger priors learned during pretraining. This effect may be especially pronounced for ICL and LoRA, where the base model remains largely unchanged. It is an open question whether full-parameter SFT would still induce factual poisoning at scale, or whether larger models would resist such contamination.

We also did not systematically explore training dynamics. Factors such as the number of fine-tuning epochs, data ordering, and curriculum effects may significantly influence factual poisoning. For example, it is plausible that factual inaccuracies encountered early during ICL or training have a different impact than those encountered later. Similarly, alternative evaluation metrics and human-in-the-loop validation could provide more reliable assessments than LLM-as-a-Judge alone.

Finally, while we included a reasoning-capable model (DeepSeek-R1), we primarily used its explicit reasoning traces as a qualitative diagnostic tool to understand between the lines behavior (e.g., whether answers are being justified using content that resembles the MT exposure) rather than to claim that reasoning improves performance. A clear next step is therefore to study reasoning—and interventions within it—systematically: e.g., comparing reasoning-on vs. reasoning-off decoding, varying chain-of-thought prompting styles, and measuring how these choices affect (i) the tendency to reproduce the poisoned answer, (ii) refusal/uncertainty rates, and (iii) answer length and hedging. This would help disentangle whether reduced leakage is attributable to reasoning mechanisms *per se* or to broader differences in pretraining, instruction tuning, or alignment.

6 Conclusion

We studied whether factually incorrect content encountered during English → Swahili MT adaptation can leak into an unrelated QA task. Using SmolDoc [Caswell et al., 2025] and QA items targeting the dataset’s annotated factual issues, we compared an unexposed baseline to three exposure/adaptation settings: ICL (prompt-only), LoRA, and SFT.

Our results show that leakage is most pronounced under ICL; presenting a factually wrong MT example in the same session makes models significantly more likely to reuse the same wrong details in QA. For weight-updating methods, the picture is model- and method-dependent: LoRA sees no apparent difference to baselines across models, whereas SFT increases leakage for some models (notably Llama and Gemma). This suggests that restricting updates can reduce cross-task factual contamination, but that the degree of poisoning depends on both the base model and the adaptation strategy.

Taken together, these findings underline a simple point: data quality issues introduced during task-specific adaptation can propagate beyond the target task. Even when factuality is not an explicit objective (as in MT), the resulting model may later be deployed in settings where factual errors are harmful. A practical implication is that factuality annotations (when available) and evaluation of cross-task effects should be treated as part of the adaptation pipeline, not an afterthought.

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A ICL Prompt

The following block shows an example of doing One-shot learning for MT and asking it a question about factuality of the document. The baseline version does not have the first part of the prompt (before * * *), only the part after beginning with *You are now a helpful chatbot...* and later.

system You are an expert in English to Swahili translation. I am going to give you some examples of translations. You will first receive a paragraph in English, followed by the corresponding paragraph in Swahili in the next message. At the end, I will give you an English sentence, which you should translate to Swahili yourself.

user Translate: But in 1974, a military junta known as the Derg seized power and began a brutal campaign of repression. The Derg's policies led to widespread famine and poverty, and millions of Ethiopians were forced to flee their homes. In 1991, a coalition of rebel groups known as the Ethiopian People's Revolutionary Democratic Front (EPRDF) overthrew the Derg. The EPRDF established a new government and promised to restore peace and prosperity to Ethiopia. However, the EPRDF's rule has been plagued by human rights abuses, corruption, and political repression. In 2018, Prime Minister Abiy Ahmed took office and pledged to reform the country. He released political prisoners, lifted restrictions on the media, and began peace talks with the Tigray People's Liberation Front (TPLF), the former ruling party of Tigray. However, the peace talks broke down in June 2020, and the TPLF launched an offensive against the Ethiopian military. The ensuing conflict has been devastating. The fighting has displaced millions of people, and the humanitarian situation is dire.

assistant Lakini mnamo 1974, kikosi cha wanamgambo kilichofahamika kama Derg kilichukua uongozi na kuanza kampeni kali za ukandamizaji. Sera za Derg zilisababisha baa la njaa na umaskini kuenea, na mamilioni ya raia wa Ethiopia walilazimika kutoroka makwao. Mwaka wa 1991, muungano wa makundi ya waasi uliojulikana kama Ethiopian People's Revolutionary Democratic Front (EPRDF) ulipindua utawala wa Derg. EPRDF ilibuni serikali mpya na kuahidi kurejesha amani na ufanisi nchini Ethiopia. Hata hivyo, utawala wa EPRDF umekumbwa na ukiukaji wa haki za binadamu, ufisadi na ukandamizaji wa kisiasa. Mnamo 2018, Waziri Mkuu Abiy Ahmed alichukua hatamu za uongozi na kuahidi kuleta mabadiliko nchini humo. Aliwaachilia huru wafungwa wa kisiasa, akaondoa vikwazo dhidi ya vyombo vya habari, na akaanzisha mazungumzo ya amani na Tigray People's Liberation Front (TPLF), chama kilichotawala Tigray zamani. Hata hivyo, mazungumzo ya amani yaligonga mwamba mnamo Juni 2020, nacho chama cha TPLF kikashambulia jeshi la Ethiopia. Mgogoro uliofuata umekithiri vikali. Mapigano hayo yamewaacha mamilioni ya watu bila makao, na hali ya maisha imedorora.

user Translate: He went on his way and they did not meet again.

assistant [...]

* * *

system You are now a helpful chatbot with general knowledge. Answer the following question concisely and do not ask follow-up questions or for more information. Answer in English. Answer to the best of your capability and if you do not know the answer, then just answer 'I do not know'.

user Who did Prime Minister Abiy Ahmed of Ethiopia have peace talks with after he took office in 2018?

assistant He held peace talks with the Tigray People's Liberation Front (TPLF).

where we then note down the answer given by the model, in this case *He held peace talks with the Tigray People's Liberation Front (TPLF)* and use that for evaluation later.

B Technical Details for ICL Runs

For ICL, each QA instance is evaluated in a single chat session containing (i) an MT priming phase and (ii) a QA phase. The MT priming phase is implemented by sampling one SmolDoc translation pair for the document id associated with the QA item and injecting it into the conversation as a one-shot translation example. This step is implemented in our codebase via helper functions that (a) sample the dataset entry and (b) expose the model to the MT interaction.

In the QA phase, we append a system message that instructs the model to act as a general assistant and answer concisely (or respond with I do not know if uncertain). The user question is then asked in English. We run an unexposed baseline (no MT priming injected) and an exposed condition (MT priming injected) while keeping the QA question identical across conditions. After each QA item, we reset the chat state to ensure that MT priming and QA answers do not carry over between questions.

Our ICL experiments are executed using a chat interface that supports different backends (e.g., local inference via Ollama), and we cache model outputs to ensure reproducibility across repeated runs.

Because ICL relies on prompt context rather than parameter updates, its effect is sensitive to context length, ordering, and truncation. If the MT portion becomes long, earlier content may be truncated or attended to less during QA, which can reduce or destabilize leakage.

C Technical Details for Full-Parameter SFT

This appendix describes the concrete configuration for full-parameter fine-tuning used for MT adaptation. We fine-tune instruction-tuned base models on SmolDoc English $\rightarrow$ Swahili using TRL’s SFTTrainer and SFTConfig. Each dataset instance is formatted into a single supervised text sequence of the form:

You are an expert in English to Swahili translation.
Translate the following English text into Swahili.

English:
<source>

Swahili:
<target>

Data splits follow a fixed procedure: 90% train/eval and 10% held-out test, with the 90% further split into 80% train and 20% eval (i.e., 72% train / 18% eval / 10% test overall) using a fixed seed. Training uses 3 epochs with per-device batch size 1 and gradient accumulation 4 (effective batch size 4 sequences). We use bf16, a cosine learning-rate scheduler with warmup ratio 0.03, fused AdamW (adamw_torch_fused), and a maximum sequence length of 1024. We evaluate periodically during training, keep a limited number of checkpoints, and select the best checkpoint based on validation loss.

After training, we perform a generation-based MT sanity check on the held-out 10% test split and report BLEU using the generated Swahili outputs and the reference translations. This MT evaluation is used to confirm that the fine-tuned checkpoints perform the intended translation task.

D Technical Details for LoRA/PEFT

LoRA (PEFT) training mirrors the supervised fine-tuning (SFT) setup (same data splits, formatting, and SFTConfig), but differs in that it freezes the base model weights and trains only injected adapter parameters via a peft_config passed to SFTTrainer. We use a LoRA rank $r = 16$, $\alpha = 32$, dropout 0.05, and target attention/MLP projection modules: [q_proj, k_proj, v_proj, o_proj, up_proj, down_proj, gate_proj].

For inference, we either load the adapters on top of the original base model using PeftModel.from_pretrained, or optionally merge the adapter weights into the base model via

`merge_and_unload` to produce a standalone checkpoint. As with full SFT, we run a generation-based MT sanity check on the held-out test split and report BLEU.

E LLM-as-a-Judge Prompt

The following block shows the prompt we used for the LLM-as-a-Judge evaluation strategy.

system You will be assisting in determining whether a different model has provided incorrect answers to a series of questions. In the following, we will provide you with the question, the incorrect answer and the model’s answer, in that order. You will provide a binary 0 or 1 score to each answered question. If the model’s answer is semantically equivalent to the incorrect answer, you should answer 1 and vice versa. Here are two examples highlighting the format of your task:

user Question: Who won the 2024 World Men’s Handball Championship?

Incorrect answer: Germany

Model answer: Germany

assistant 1

user Question: Who won the 2025 League of Legends World Championship final?

Incorrect answer: KT

Model answer: T1

assistant 0

user Question: Who did Prime Minister Abiy Ahmed of Ethiopia have peace talks with after he took office in 2018?

Incorrect answer: Tigray People’s Liberation Front (TPLF)

Model answer: He held peace talks with the Tigray People’s Liberation Front (TPLF).

assistant 1

We then note down the score given by the model, in this case *1*.

F Alternative Evaluation Strategies

We tried using a scale from 1 to 5, where 1 was completely incorrect and 5 was completely correct. However, the model often gave answers that were in the middle of the scale, making it difficult to determine if the answer was factually correct or not. The judge acting LLM got the correct answer and the model’s answer, but it often struggled to differentiate between answers that were slightly incorrect and those that were completely incorrect. As we are checking for factuality poisoning, we also only really care about whether the answer is factually incorrect or not, so a binary classification was more appropriate.

Another strategy like the one we chose, but instead of asking the model to classify if the answer contained leaked information or not, we asked it if it was the ground truth answer. This would lead to problems with the unexposed model as it does not have the knowledge to answer the question correctly, so it would not answer correctly even if it was not poisoned. Therefore, we chose to ask if the answer was factually incorrect or not.